

Dr. Abhinav Jha

Date of Birth: 2nd July 1994 ◊ **Place of Birth:** New Delhi, India

Current Address: Indian Institute of Technology, Gandhinagar, Gujarat ◊ abhinav.jha@iitgn.ac.in

Research Interests

Numerical Analysis of Partial Differential Equations, Stabilization Methods for Convection Dominated Problems, A Posteriori Error Estimates, Domain Decomposition Methods in Computational Chemistry, Scientific Computing, including writing scientific software.

Work Experience

Indian Institute of Technology, Gandhinagar

October 2024 - Present

Assistant Professor, Grade 1

Department of Mathematics

Universität Stuttgart, Stuttgart

September 2022 - August 2024

Postdoctoral Researcher, Mathematics.

Group: Numerical Mathematics for High Performance Computing

Advisor: Prof. Dr. Benjamin Stamm.

RWTH Aachen University, Aachen

January 2021 - August 2022

Postdoctoral Researcher, Mathematics.

Group: Applied and Computational Mathematics

Advisor: Prof. Dr. Benjamin Stamm.

Education

Freie Universität, Berlin

October 2017 - October 2020

PhD, Mathematics.

Grade: Magna cum Laude

Title: Numerical Algorithms for Algebraic Stabilizations of Scalar Convection-Dominated Problems.

Advisor: Prof. Dr. Volker John.

Indian Institute of Technology, Roorkee

July 2015 - July 2017

Master of Science, Mathematics.

CGPA: 9.59/10.0

Title: Finite Element Method for Population Balance Equations.

Advisor: Prof. Dr. Ankik Kumar Giri.

St. Stephen's College, University of Delhi

July 2012 - July 2015

Bachelor of Science, Mathematics.

Overall Percentage: 89.3%

Publications

Published

1. Naveed Ahmed and **Abhinav Jha**: *Adaptive Finite Elements with Algebraic Stabilization for Convection-Dominated Transport*. Journal of Computational Physics, (**accepted**), 2026.
2. Thiago Carvalho Corso, Muhammad Hassan, **Abhinav Jha**, and Benjamin Stamm: *Trace estimates for harmonic functions along circular arcs with applications to domain decomposition on overlapping disks*. SIAM Journal on Mathematical Analysis, **58**, 1571-1612, 10.1137/24M1710310, 2026, **Q1**, IF: 1.9.
3. Naveed Ahmed, Samir Karaa, and **Abhinav Jha**: *Symmetric Stabilized FEM for Time-Fractional Convection-Diffusion-Reaction Equations*. Mathematics and Computers in Simulation, **245**, 685-697, 10.1016/j.matcom.2026.02.003, 2026, **Q1**, IF: 4.4.
4. Petr Knobloch, Dmitri Kuzmin, and **Abhinav Jha**: *Well-balanced convex limiting for finite element discretizations of steady convection-diffusion-reaction equations*. Journal of Computational Physics, **518**, 113305, 10.1016/j.jcp.2024.113305, 2024, **Q1**, IF: 3.8.

5. Michele Nottoli, Michael F. Herbst, Aleksandr Mikhalev, **Abhinav Jha**, Filippo Lipparini, and Benjamin Stamm: *ddX: Polarizable Continuum Solvation from Small Molecules to Proteins*. WIREs Computational Molecular Science, **14**, e1726, 10.1002/wcms.1726, 2024, **Q1**, IF: 27.0.
6. **Abhinav Jha**: *Residual-based a posteriori error estimators for algebraic stabilizations*. Applied Mathematics Letters, **157**, 109192, 10.1016/j.aml.2024.109192, 2024, **Q1**, IF: 2.8.
7. **Abhinav Jha**, Volker John, and Petr Knobloch: *Adaptive Grids in the Context of Algebraic Stabilizations for Convection-Diffusion-Reaction Equations*. SIAM Journal on Scientific Computing, **45**, B564-589, 10.1137/21M1466360, 2023, **Q1**, IF: 2.6.
8. **Abhinav Jha**, Michele Nottoli, Aleksandr Mikhalev, Chaoyu Quan, and Benjamin Stamm: *Linear scaling computation of forces for the domain-decomposition linear Poisson–Boltzmann method*. The Journal of Chemical Physics, **150**, 104105, 10.1063/5.0141025, 2023, **Q1**, IF: 3.1.
9. **Abhinav Jha**, Ondřej Pártl, Naveed Ahmed, and Dmitri Kuzmin: *An Assessment of Solvers for Algebraically Stabilized Schemes applied to Convection Diffusion Reaction Equations*. Journal of Numerical Mathematics, **31**, 79-103, 10.1515/jnma-2021-0123, 2023, **Q1**, IF: 4.0.
10. **Abhinav Jha**: *Hanging Nodes for Higher-Order Lagrange Finite Elements*. Examples and Counterexamples, **1**, 100025, 10.1016/j.exco.2021.100025, 2021, **Q4**, IF: 0.7.
11. **Abhinav Jha**: *A Residual Based A Posteriori Error Estimators for AFC Schemes for Convection-Diffusion Equations*. Computer and Mathematics with Applications, **97**, 86-99, 10.1016/j.camwa.-2021.05.031, 2021, **Q1**, IF: 2.5.
12. **Abhinav Jha** and Volker John: *On basic iteration schemes for nonlinear AFC discretizations*. Boundary and Interior Layers, Computational and Asymptotic Methods BAIL 2018, **135**, 113–128, 10.1007/978-3-030-41800-7_7, 2020, **QNA**, IF: NA.
13. **Abhinav Jha** and Volker John: *A Study of Solvers for Nonlinear AFC Discretizations of Convection-Diffusion Equations*. Computer and Mathematics with Applications, **78**, 3117-3138, 10.1016/j.camwa.2019.04.020, 2019, **Q1**, IF: 2.5.

Submitted

1. Sarthak Sourav Dash and **Abhinav Jha**: *Bilinearity Preserving Algebraic Flux Correction*. [arXiv].
2. Pinki Khatun, M. Sajid, **Abhinav Jha**, and M. Tanveer: *Physics-Informed Broad Learning System: An Efficient Backpropagation-Free Framework for Solving Partial Differential Equations*. [arXiv].
3. Ajeet Singh and **Abhinav Jha**: *Lyapunov Stability and Optimal Error Estimates for an SIPG Method for Weakly Damped Semilinear Wave Equations*. [arXiv].
4. Harshit Bajpai, Ankik Kumar Giri, Tim Jahn, and **Abhinav Jha**: *On the convergence of an adaptive denoise driven iterative regularization with early stopping*. [arXiv].
5. **Abhinav Jha**, Samir Karaa, and Aditi Tomar: *Non-uniform α -Robust Alikhanov Mixed FEM with Optimal Convergence for the Time-Fractional Allen–Cahn Equation*. [arXiv].
6. **Abhinav Jha** and Benjamin Stamm: *Domain decomposition method for Poisson–Boltzmann equations based on Solvent Excluded Surface*. [arXiv].

Under Preparation

1. Naveed Ahmed, Lourenco Beirao Da Veiga, Franco Dassi, and **Abhinav Jha**: *Algebraic Stabilizations in the context of Virtual Element Methods*.

Grants

Granted

- **Advances in Domain Decomposition Methods for Implicit Solvation Models**
 - Prime Minister Early Career Research Grant
 - Funding Agency: *ANRF*.
- **Algebraic Stabilization Schemes for Evolutionary Convection-Diffusion Equation**
 - DST INSPIRE Faculty Award
 - Funding Agency: *Department of Science and Technology*.

· **A Posteriori Error Estimation for Flux Corrected Transport Schemes**

- Institute Research Grant
- Funding Agency: *Indian Institute of Technology, Gandhinagar.*

Under Review

· **Further Advancements of Galerkin Neural Network Methods**

- Funding Agency: *National Board for Higher Mathematics.*

Presentation in Conferences

- *Domain Decomposition Methods for Implicit Solvation Models*, Workshop on Advanced Methods in Integral Equations and Their Applications (WAMIEA 2026), 17th – 19th August 2026, IIT Jodhpur, Rajasthan, **[invited talk]**.
- *Recent Advancements in a Posteriori Error Estimation for Algebraic Stabilizations*, 17th World Congress on Computational Mechanics & 10th Asian Pacific Congress on Computational Mechanics, 19th – 24th July 2026, Munich, Germany.
- *Advanced Topics in Finite Element Methods*, NAFED 09, 25th – 26th May 2026, Vikram Sarabhai Space Center, Gujarat, **[invited talk]**.
- *Recent Advancements in a Posteriori Error Estimation for Algebraic Stabilizations*, ENUMATH 2025, 1st September – 5th September 2025, Heidelberg, Germany.
- *Domain Decomposition Methods for the Poisson–Boltzmann Equations*, 93rd Annual Meeting of the International Association of Applied Mathematics and Mechanics, 30th May – 2nd June 2023, Dresden, Germany.
- *Adaptive Grids for Algebraic Stabilizations of Convection-Diffusion-Reaction Equations*, SIAM Conference on Computational Science and Engineering (CSE23), 26th February – 3rd March 2023, Amsterdam, Netherlands.
- *Computation of Forces Arising from the Linear Poisson-Boltzmann Method in the Domain Decomposition Paradigm*, 92nd Annual Meeting of the International Association of Applied Mathematics and Mechanics, 15th – 18th August 2022, Aachen, Germany.
- *A Residual based a Posteriori Error Estimators for Algebraic Flux Correction Scheme*, 15th World Congress on Computational Mechanics & 8th Asian Pacific Congress on Computational Mechanics, 31st July – 5th August 2022, Yokohoma, Japan.
- *Adaptive Grids for Algebraic Stabilizations of Convection-Diffusion-Reaction Equations*, Workshop on Numerical Methods and Analysis in CFD, 5th – 8th July 2022, WIAS, Berlin, Germany.
- *Adaptive Grids for Algebraic Stabilizations of Convection-Diffusion-Reaction Equations*, 18th Workshop on Numerical Methods for Problems with Layer Phenomena, 24th – 26th March 2022, Hagen, Germany.
- *Residual based a Posteriori Error Estimators for Algebraic Flux Correction Scheme*, Chemnitz Finite Element Symposium 2021, 6th – 8th September 2021, Online.
- *Residual based a Posteriori Error Estimators for Algebraic Flux Correction Scheme*, Bound-Preserving Space and Time Discretizations for Convection-Dominated Problems, BIRS & CMO, 22nd – 27th August 2021, Online, **[invited talk]**.
- *Towards A Posteriori Error Estimators for Algebraic Flux Correction Scheme*, ESCO 2020, 7th International Congress of Computational Engineering and Sciences, 8th – 12th June 2020, Online.
- *On Numerical Simulations and a Posteriori Analysis for Algebraic Flux Correction Schemes*, MAFE-LAP 2019, The Mathematics of Finite Elements and Applications 2019, 17th – 21st June 2019, Brunel University, London.
- *On Numerical Simulations and a Posteriori Analysis for Algebraic Flux Correction Schemes*, The 28th Biennial Numerical Analysis Conference, 25th – 28th June 2019, University of Strathclyde, Glasgow.

- *Investigation of different solvers for nonlinear algebraic stabilizations of convection diffusion equations*, 13th International Workshop on Variational Multiscale and Stabilized Finite Elements, 5th – 7th December 2018, Weierstrass Institute for Applied Analysis and Stochastic, Berlin.
- *Study of Iterative Methods for Nonlinear AFC Discretizations on Convection-Diffusion Equations*, BAIL 2018, International Conference on Boundary and Interior Layers, 18th – 22nd June 2018, Glasgow, Scotland.

Organisation of Conferences

- *Minisymposium: Advances in Numerical Methods for Problems from Fluid Mechanics*, ENUMATH 2025, 1st September – 5th September 2025, Heidelberg, Germany.
- *Minisymposium: Special Methods in Computational Fluid Mechanics*, 15th World Congress on Computational Mechanics & 8th Asian Pacific Congress on Computational Mechanics, 31st July – 5th August 2022, Yokohoma, Japan.
- 8th *BMS Student Conference*, 19th – 22nd February 2020, Technische Universität, Berlin.

Research Visits

- Prof. Dr. Filippo Lipparini, Department of Chemistry and Industrial Chemistry, Università di Pisa, Italy; 27th – 30th March 2022.
- Prof. Dr. Benjamin Stamm, Applied and Computational Mathematics, RWTH Aachen University, Germany; 10th November – 24th December 2020.

Software Developed

ddPB-SES - Domain Decomposition for Poisson-Boltzmann Equation *Language: MATLAB*
Universität Stuttgart, Stuttgart

- Developed the Domain Decomposition method for Poisson Boltzmann Equation on SES cavity.
- Solves both the Linear and Non-Linear equation.

ddX - Domain Decomposition Paradigm for Continuum Solvation Models *Language: Fortran-90*
RWTH Aachen University, Aachen

- Developed the Domain Decomposition Linear Poisson Boltzmann (ddLPB) sub-module.
- Co-developed the general framework of the package.

ParMooN - Parallel Mathematics and object-oriented Numerics *Language: C++*
Weierstraß Institute for Applied Analysis and Stochastic, Berlin

- Developed the Algebraic Flux Correction package for Steady-State and Time-Dependent Convection-Diffusion Equations.
- Co-developed the a Posteriori Estimator package.

Online Course Development

NumLinPy: A Workshop on the Application of Numerical Linear Algebra in Python 10th June 2026
NPTEL+ **Evaluation:** 4.56/5

- 138 Participants

NumPython 2: A Workshop on Numerical Analysis of ODEs 16th – 17th May 2026
NPTEL+ **Evaluation:** 4.64/5

- 74 Participants

NumPython: A Workshop on Numerical Analysis 7th – 8th February 2026
NPTEL+ **Evaluation:** 4.5/5

- 265 Participants

Teaching Duties

Indian Institute of Technology, Gandhinagar

- Instructor for *MA103: Calculus of Single Variable*, Semester 1, 2026.
Evaluation: Ongoing
- Instructor for *MA643: Numerical Analysis of PDEs*, Semester 1, 2026.
Evaluation: Ongoing
- Instructor for *MA103: Calculus of Single Variable*, Summer Semester, 2026.
Evaluation: NA
- Instructor for *MA203: Numerical Methods*, Semester 2, 2026.
Evaluation: 3.45/4
- Instructor for *MA643: Numerical Analysis of PDEs*, Semester 1, 2025.
Evaluation: 3.86/4
- Instructor for *MA637: Numerical Analysis and Computing*, Semester 2, 2025.
Evaluation: 3.57/4
- Tutor for *MA205: Calculus of Several Variables*, Semester 1, 2025.
- Tutor for *MA203: Numerical Methods*, Semester 2, 2025.
Evaluation: 3.5/4

Univeristät Stuttgart

- TA for *Numerische Grundlagen für ernen, fmt, mach, mawie*, Summer Semester, 2024, 2023.
- TA for *Numerische Mathematik 1*, Winter Semester, 2023 .
- TA for *Numerical Methods for Differential Equations*, Summer Semester, 2023 .
- Supervisor for *Hauptseminar Numerische Analysis und Simulation*, Winter Semester, 2022 .
- TA for *Höhere Mathematik 3 für Ingenieurstudiengänge*, Winter Semester, 2022 .

RWTH Aachen University

- TA for *Mathematische Grundlagen II (CES)*, Summer Semester, 2022 .
- TA for *Partial Differential Equations (CES+SiSc)*, Winter Semester, 2021 .
- TA for *Mathematische Grundlagen IV (CES)*, Summer Semester, 2021 .

Freie Universität Berlin

- TA for *Numerical Methods for PDEs -Numerik III*, Summer Semester, 2019 .
- TA for *Numerical Methods for ODEs and Numerical Linear Algebra-Numerik II*, Winter Semester, 2018 .

*TA = Teaching Assistant

Supervision

Sarthak Sourav Dash

Indian Institute of Technology, Gandhinagar

July 2025 –
PhD

- Research Work: Adaptive Finite Element for Parabolic Problems

Ajeet Singh

Indian Institute of Technology, Gandhinagar

May 2026 – September 2026
PostDoc

- Current Position: PostDoc at Indian Institute of Technology, Delhi
- Research Work: Discontinuous Galerkin Methods.

Pinki Khatun

Indian Institute of Technology, Gandhinagar

May 2025 – October 2025
PostDoc

- Current Position: PostDoc at University of Florence, Italy
- Research Work: Physics Informed Embedded Neural Network

Aditi Tomar

Indian Institute of Technology, Gandhinagar

May 2025 – December 2025

PostDoc

- Current Position: PostDoc at TU Wien, Vienna
- Research Work: Time-Fractional Differential Equation

Junghoon Lee

Universität Stuttgart

April 2023 – October 2023

Masters

- Research Work: A Posteriori Error Estimators for Laplace Eigenvalue Problems.

Certifications

Machine Learning Specialisation

Coursera

November 2023

Projects and Fellowships

Finite Element Method for Population Balance Equation

Indian Institute of Technology Roorkee, India

January 2017 - May 2017

- Developed Convergence Analysis of Finite Element Method (Collocation Method) for Population Balance Equations.

Professor Nagpaul Fellowship

St. Stephen's College, University of Delhi

October 2014 - May 2015

- Researched on Network Optimization and its applications in daily life.

Summer Research Fellowship

Indian Institute of Science Bangalore, India

May 2014 - June 2014

- Derived continuous time domain representation of Riesz Transform in two dimensions using Fourier transforms.

Position of Responsibility

Indian Institute of Technology, Gandhinagar

May 2025-

- Faculty Advisor for M.Sc. Mathematics 2028 Batch.
- Web Administrator of the Department Website.
- Co-organizer of the Foundation Program 2025.

Berlin Mathematical School, Berlin

Student Representative

December 2018 - December 2019

- Member of the Executive board and the Admissions Committee.
- Organized the Career Event 2019.
- Organized the 8th BMS Student Conference.

The Mathematics Society, St. Stephen's College

President

July 2014 - July 2015

- Initiated the Professor Nagpaul Fellowship.
- Initiated the Professor Mathur Memorial Lecture Series.
- Editor of Society Magazine, *Mathematica*.
- Organized *MathSoc Open 2014* and *MathSoc Open 2015*.

Gandhi Study Circle, St. Stephen's College
Vice President

July 2014 - July 2015

- Coordinated the Regional Study Conference, August 2013.
- Member of the organizing team that held Mock Parliament, February 2014.

Scholarships and Awards

DST INSPIRE Faculty Award

June 2025 - June 2030

Ministry of Human Resources and Development, India.

BMS Phase 2 Scholarship

October 2017 - September 2020

Berlin Mathematical School.

Dr. Gorakh Prasad Scholarship

July 2015 - July 2017

Indian Institute of Technology, Roorkee.

1st Rank in Department

INSPIRE Scholarship

July 2012 - July 2017

Ministry of Human Resources and Development, India.

Department of Mathematics Leadership Award

April 2015

St. Stephen's College, University of Delhi.

Kesar Devi Scholarship

April 2013

St. Stephen's College, University of Delhi.

National Examinations

CSIR-UGC NET JRF

June 2016

Rank : 28.

Joint Application test for M.Sc. (JAM)

February 2015

Rank : 196.

Technical Strengths

Programming Language	C++, Fortran
Scripting Language	Python
Operating System	Linux, MacOS
Version Control	Mercurial, Git
Software & Tools	Mathematica, Matlab, L ^A T _E X, Photoshop CS5

Reviewer for Journals

Journal of Computational and Applied Mathematics	<i>Elsevier</i>
International Journal of Computational Mathematics	<i>Taylor & Francis</i>
SIAM Journal on Numerical Analysis	<i>SIAM</i>
MathSciNet	<i>AMS</i>
The Journal of Supercomputing	<i>Springer</i>
Mathematical Control and Related Fields	<i>AIMS</i>
Journal of Numerical Mathematics	<i>De Gruyter</i>
Multidiscipline Modeling in Materials and Structures	<i>Emerald Group</i>
Journal of Chemical Theory and Computation	<i>ACS Publications</i>
Zeitschrift für angewandte Mathematik und Physik	<i>Springer</i>
Computer Physics Communications	<i>Elsevier</i>

References

Prof. Dr. Volker John

john@wias-berlin.de

Doctoral Supervisor

- Freie Universität, Berlin & Weierstrass Institute for Applied Analysis and Stochastics.

Prof. Dr. Benjamin Stamm

best@ians.uni-stuttgart.de

Postdoctoral Supervisor

- Universität Stuttgart, Stuttgart.

Prof. Dr. Petr Knobloch

knobloch@karlin.mff.cuni.cz

Research Collaborator

- Charles University, Prague.